

Vadim Aliautdinov

Senior GIS Specialist | ArcGIS Enterprise | Spatial Data Systems | Remote Sensing & ML

Portfolio: <https://vadagas.github.io> | GitHub: <https://github.com/vadagas>

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PERSONAL STATEMENT

Senior GIS Specialist with 15+ years of experience delivering enterprise and national-scale geospatial solutions for government and commercial organisations. Owns the full stack: ArcGIS Enterprise administration (99%+ uptime, zero unplanned downtime across 9 years), Python/FME automation (~20 hrs/week of manual work eliminated), 4 TB+ LiDAR/raster pipelines, and ML-based national mapping products — including the public cadastral map of Russia and reference forest maps validated at 85.8–87.3% overall accuracy. Seeking a Senior GIS Engineer or GIS Technical Lead role.

KEY SKILLS

- **Enterprise Administration:** multi-tier ArcGIS Enterprise (Portal, Server, Data Store), ArcGIS Online; 99%+ uptime and zero unplanned downtime across 9 years; 15–30 concurrent client environments as primary technical authority.
- **Automation & Programming:** Python (ArcPy, GeoPandas, Pandas) and SQL for ETL and geoprocessing automation; cut data ingestion by ~60% and eliminated ~20 hrs/week of manual work.
- **Data Integration & ETL:** FME-based scalable workflows; complex spatial databases (PostgreSQL/PostGIS, MS SQL, Oracle); index and query optimisation (2–3× faster production services).
- **Advanced Data Processing:** 4 TB+ LiDAR, DEM and raster pipelines (tile generation: days → hours); 1,120+ layers consolidated into an 80 GB geodatabase.
- **Remote Sensing & Machine Learning:** Random Forest regression/classification and maximum-likelihood classification over MODIS / Landsat / Sentinel-2 seasonal composites with state forest inventory (GIL) ground data and FABDEM relief; national forest-characteristic layers validated at 85.8–87.3% overall accuracy (R^2 up to 0.83).
- **Web GIS & Dashboards:** interactive Web GIS applications and operational dashboards for monitoring and decision support.
- **Vector & Raster Data Processing:** topology editing, feature generalisation, spatial joins, overlay analysis, multi-band raster processing (clipping, mosaicking, reprojection, classification) in ArcGIS Pro, QGIS, GDAL/OGR.
- **Coordinate Systems & Projections:** geodetic datums and CRS (WGS84, ETRS89, Pulkovo 1942/SK-95, UTM, Gauss-Krüger), datum transformations, on-the-fly reprojection in enterprise environments.
- **Cartographic Design & Map Production:** thematic and topographic design, symbology standardisation, map series automation, print-ready layouts to national standards; unified multi-scale national base maps.

PROFESSIONAL EXPERIENCE

Senior GIS Specialist — Institute of Geographic Information Technologies (IGIT) | 2023 – Present

- Administers a multi-tier ArcGIS Enterprise production environment (Portal · Server · Data Store) underpinning 3 national government programmes with zero unplanned downtime.
- Built Python/FME pipelines consolidating 1,120+ layers and an 80 GB geodatabase, eliminating ~20 hrs/week of manual data preparation.
- Developed scalable LiDAR/raster pipelines for 4 TB+ datasets, cutting tile generation from days to hours and enabling near-real-time analyst access.
- Delivered national-scale geospatial products: ML-modelled reference forest maps of the Russian Federation (85.8–87.3% overall accuracy) and the public cadastral map with a unified multi-scale base map.
- Optimised spatial query performance 2–3× across production services via index redesign and query-plan optimisation.
- Empowers 10+ government analysts through structured training and technical documentation, driving GIS adoption across 3 departments.

Senior GIS Specialist / ArcGIS Enterprise Specialist — Data+ (Esri Distributor) | 2014 – 2023

- Architected and delivered ArcGIS Enterprise solutions for 15–30 concurrent government and commercial environments as primary technical authority for Esri's largest regional distributor.
- Reduced data ingestion time by ~60% across 10+ client projects by replacing manual workflows with Python & FME ETL pipelines.
- Maintained 99%+ uptime across a portfolio of production ArcGIS Server environments over 9 years, retaining all key accounts through expert-level incident response.
- Mentored 3 junior GIS specialists into independent practitioners through structured code review and hands-on mentoring, halving team onboarding time.

GIS Specialist — ERM Eurasia Limited | 2013 – 2014

- Engineered spatial database schemas for 5+ environmental & geophysical projects, establishing data-integrity standards adopted across regional workflows.
- Automated data-loading validation routines, eliminating manual QA checks and accelerating environmental report delivery by ~30%.

GIS Researcher — State Oceanographic Institute | 2011 – 2013

- Developed a publicly accessible web marine atlas and bespoke spatial clustering and pollutant-dispersion modelling tools for marine researchers.

SELECTED PROJECTS

- **Public Cadastral Map of Russia:** full-cycle development — processing of all spatial data, ongoing technical support, and a unified national base map published across 6 scale levels.
- **Reference Forest Maps of the Russian Federation:** ML modelling (Random Forest, maximum-likelihood) of forest area, dominant species, age, growing stock, stand density, canopy height and site class from MODIS/Landsat/Sentinel-2 composites, GIL ground data and FABDEM; 925 M ha mapped, 106.8 B m³ of growing stock estimated; validated at 85.8–87.3% overall accuracy.
- **Enterprise GIS Platforms:** robust ArcGIS Enterprise environments improving reliability of regional spatial data access.

- **Flood Modelling & Risk Analysis:** geoprocessing workflows for environmental risk assessment, supporting emergency management and Civil Defence readiness.
- **Web GIS Solutions:** dashboards and interactive maps translating complex spatial data into operational insights.

EDUCATION & QUALIFICATIONS

- Postgraduate Research – Geoecology, State Oceanographic Institute.
- Master's Degree – Geography, Cartography & Geoinformatics, Lomonosov Moscow State University.

CERTIFICATIONS

- Esri Technical Certification: ArcGIS Pro Associate (2022).
- ArcGIS Enterprise: Administration & Configuration (2023).
- Spatial Analysis with ArcGIS Pro | Building Geodatabases.

INTERESTS Travel, volleyball, table tennis, and yachting — sailing ties back to the oceanographic-research background; team sports keep collaboration and quick decision-making sharp.